The Coding Interview Patterns Playbook

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Course Introduction

Requirements

- Basic knowledge of any programming language - ideally Kotlin
- Optional: Understanding of the Gradle build system*
- **Preinstalled on computer:**
  - IntelliJ IDEA (Community Edition)**
  - Java 17***

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*http://udemy.com/course/mastering-gradle/

*https://www.jetbrains.com/idea/download/?section=mac

**https://adoptium.net/en-GB/temurin/releases/?version=17&os=mac&arch=aarch64&package=jre

Welcome to the Course

- Why "Patterns" instead of "Problems"?
- Moving from memorization to intuition
- The 80/20 Rule: 15 patterns solve 80% of interviews

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How to Use This Course Effectively

- **Course Approach:** We will start by showing the brute-force solution, and then transition to the optimal solution.
- **In an Interview (Ideal Case):**
  1. **Identify:** Recognize pattern triggers in the problem text
  2. **Trace:** Dry-run the logic on a small input
  3. **Implement:** Code the solution without looking at hints

Interview Mindset: Thinking in Patterns

- Stop looking at the "Story" (e.g., "Alice has 3 apples...")
- Start looking at the **Data Flow**

Problem Trait	**Pattern Strategy**
Finding a sub-segment in a list	Sliding Window
Searching a sorted range	Binary Search
Shortest path in unweighted graph	BFS

Common Interview Mistakes

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- **Mistake 2:** When stuck: avoiding brute force and trying to guess optimal solution
- **Mistake 3:** Forgetting to state Big O complexity
- **Mistake 4:** Not testing for $n = 0$ or $n = 1$
- **Mistake 5:** Not listening to the interviewer

Correction: Think out loud. Your interviewer is your collaborator, not your judge.

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Course Structure

- Fundamentals & Big O
- Section 3: Two Pointers Pattern
- Section 4: Sliding Window
- Section 5: Monotonic Stack
- Section 6: Prefix Sum & Cumulative
- Section 7: Cyclic Sort
- Section 8: Merge Intervals
- Section 9: Modified Binary Search
- Section 10: Fast & Slow Pointers
- Section 11: Tree Depth First Search (DFS)
- Section 12: Tree Breadth First Search (BFS)
- Section 13: Matrix Traversal (Islands)
- Section 14: Union-Find (Disjoint Set)
- Section 15: Top 'K' Elements (Heap)
- Section 16: Dynamic Programming
- Section 17: Backtracking & DFS
- Section 18: Greedy Algorithms
- Section 19: Data Streams & Online
- Section 20: Advanced Patterns
- Section 21: Miscellaneous Techniques
- Section 22: Summary & SDLC

Fundamentals & Big O

The Absolute Essentials

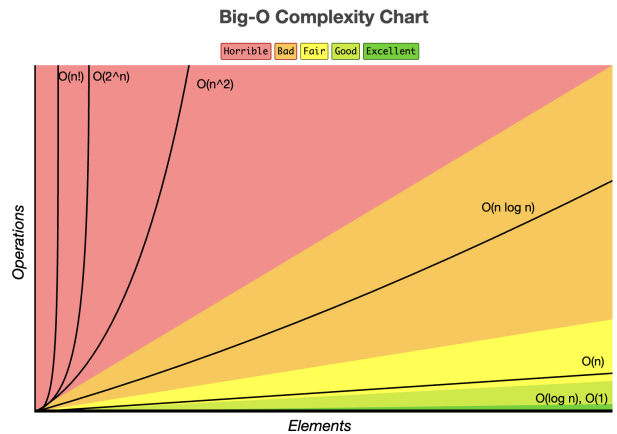
Before mastering coding patterns, you must master the fundamental building blocks.

- **Why?** Interviewers evaluate your ability to select the right tool for the job.
- **Key Areas:**
  - Analyzing efficiency (Big O Complexity).
  - Choosing the right data structures.
  - Writing idiomatic, clean code under pressure.

Big O Notation Demystified

How does the runtime or space requirement grow as input size N increases?

- **Time Complexity:** Count dominant operations.
  - $O(1)$ - Constant time.
  - $O(\log N)$ - Logarithmic (halving search space).
  - $O(N)$ - Linear (single pass).
  - $O(N \log N)$ - Linearithmic (sorting).
  - $O(N^2)$ - Quadratic (nested loops).
- **Space Complexity:** Measure auxiliary memory (call stack, maps, arrays).



bigocheatsheet.com

Growth Rates in Action ($N = 1,000,000$)

What do these complexities mean in practice for 1 million input elements?

Complexity	**Operations ($N = 10^6$)**	**Approx. Time (1GHz CPU)**
$O(1)$	1	$\approx$ 1 ns (Instant)
$O(\log N)$	≈ 20	$\approx$ 20 ns (Instant)
$O(N)$	$1,000,000$	$\approx$ 1 ms (Instant)
$O(N \log N)$	$\approx 2 \times 10^7$	$\approx$ 20 ms (Instant)
$O(N^2)$	10^{12}	$\approx$ 16.7 minutes (**Timeout**)
$O(2^N)$	$2^{1,000,000}$	$\gg$ Age of the Universe (**Impossible**)

Data Structures Crash Course

Data Structure	Strengths	Common Use Case
Array / List	Fast index access ($O(1)$)	Ordered traversal, sorting
HashMap / Map	Key lookup ($O(1)$ average)	Frequency counting, cache
HashSet / Set	Duplicate removal, search ($O(1)$)	Visited trackers, unique items
Queue / Deque	FIFO ordering ($O(1)$ push/pop)	BFS, level order traversal

Writing Idiomatic Code under Pressure

Write production-ready, readable code rather than terse competitive code.

- **Kotlin Idioms:**
  - Use ‘maxOf(a, b)‘ and ‘minOf(a, b)‘ instead of custom ifs.
  - Prefer ‘when‘ expressions for clean decision branches.
  - Avoid force-unwrapping nullables (‘!¡) unless absolutely safe.
- **Self-Documentation:** Use descriptive variable names (‘left‘, ‘right‘, ‘charCount‘) instead of obscure ones (‘l‘, ‘r‘, ‘c‘).

Section 3 — Two Pointers Pattern

What is the Two Pointers Pattern?

- **The Core Idea:** Use two indices (pointers) to iterate through a data structure simultaneously.
- **Why it works:** It reduces $O(n^2)$ nested loops to a single $O(n)$ pass.
- **Common Triggers:**
  - The array is **Sorted**.
  - You need to find a pair, a triplet, or a subarray.
  - You need to "filter" or "rearrange" an array in-place.

Pattern Mechanics: Opposite Ends

When searching for a target in a **Sorted** array:

1. **Left Pointer:** Starts at index 0.
2. **Right Pointer:** Starts at index $n - 1$.
3. **Decision:**
   - If $sum > target$: Move Right pointer left ($\leftarrow$) to decrease sum.
   - If $sum < target$: Move Left pointer right ($\rightarrow$) to increase sum.

Problem 1: 3Sum

Problem 2: Container With Most Water

Problem 3: Remove Duplicates from Sorted Array

Section 4 — Sliding Window

What is the Sliding Window Pattern?

- **The Core Idea:** Maintain a contiguous window over a linear data structure (array/string) that dynamically expands or contracts.
- **Why it works:** Converts nested $O(n^2)$ loops searching subarrays into a single $O(n)$ pass by reusing computed information.
- **Common Triggers:**
  - Subarray or substring problems.
  - Finding shortest, longest, or target length matching constraint.
  - Tracking unique elements/frequencies in a range.

Dynamic vs. Fixed Window

- **Fixed Window:** Size K is constant. Move window by adding next element and removing the leftmost.
- **Dynamic Window:** Size changes.
  - Expand the window by moving ‘right‘ pointer.
  - If constraint is violated, contract window by moving ‘left‘ pointer until constraint is satisfied again.

Problem 1: Longest Substring Without Repeating Characters

Problem 2: Permutation in String

Problem 3: Fruit Into Baskets

Problem 4: Minimum Window Substring

Section 5 — Monotonic Stack

What is a Monotonic Stack?

- **The Core Idea:** A stack that maintains elements in a strictly increasing or decreasing order.
- **How it works:** When a new element is processed, we pop all elements from the stack that violate the monotonic property before pushing the new element.
- **Common Triggers:**
  - Finding the "next greater" or "previous smaller" element in an array.
  - Computing areas bounded by heights (e.g. histograms, trapping water).
  - Resolving queries in $O(N)$ total time where each index is pushed/popped at most once.

Monotonic Stack: Types

- **Monotonic Increasing Stack:** Stack elements are sorted in ascending order from bottom to top (e.g. ‘[1, 3, 5, 8]‘). Useful to find the first smaller element to the left/right.
- **Monotonic Decreasing Stack:** Stack elements are sorted in descending order from bottom to top (e.g. ‘[9, 7, 4, 2]‘). Useful to find the first larger element to the left/right.

Problem 1: Daily Temperatures

Problem 2: Next Greater Element I

Problem 3: Trapping Rain Water

Problem 4: Largest Rectangle in Histogram

Section 6 — Prefix Sum & Cumulative Patterns

What is the Prefix Sum Pattern?

- **The Core Idea:** Precompute cumulative sums of an array: ‘prefix[i] = nums[0] + ... + nums[i]‘.
- **Why it works:** Allows you to calculate the sum of any subarray ‘nums[i...j]‘ in $O(1)$ time:

$$\text{Sum}(i, j) = \text{prefix}[j] - \text{prefix}[i - 1]$$

- **Common Triggers:**
  - Range sum query problems with static arrays.
  - Finding subarrays that sum to a target value K.
  - Subarrays with properties related to divisibility or modulo arithmetic.

Combining Prefix Sum with HashMap

To find subarrays summing to K in $O(N)$ time:

- Maintain running cumulative sum ‘currentSum‘.
- At each step, we look for a previous prefix sum that equals ‘currentSum - K‘.
- Use a HashMap to store frequencies of prefix sums seen so far.
- **Key Modulo Trick:** For divisibility by K, store ‘currentSum

Problem 1: Subarray Sum Equals K

Problem 2: Subarray Sums Divisible by K

Section 7 — Cyclic Sort

What is the Cyclic Sort Pattern?

- **The Core Idea:** When an array contains numbers in a range from 1 to n (or 0 to n), we can sort it in $O(n)$ time and $O(1)$ auxiliary space.
- **How it works:** Iterate through the array. If the current number ‘nums[i]‘ is not at its correct index (i.e. ‘nums[i] != i + 1‘), swap it with the number at its correct index ‘nums[nums[i] - 1]‘. Repeat this until the correct number is at index ‘i‘, then move to the next index.
- **Common Triggers:**
  - Problems dealing with numbers in a continuous range.
  - Finding duplicate or missing numbers in an array.
  - Finding the first/smallest missing positive number.

Cyclic Sort Mechanics

- **Index Rule:** A value ‘val‘ belongs at index ‘val - 1‘ (or index ‘val‘ if range is 0 to n).
- **Loop condition:** We only swap when the target index has a different value:

$$\text{nums}[i] \neq \text{nums}[\text{nums}[i] - 1]$$

- **Complexity analysis:** Although we have a nested loop or repeat swaps at the same index, each swap puts at least one number in its correct position. Thus, at most $N - 1$ swaps are performed, leading to $O(N)$ time.

Problem 1: Find All Numbers Disappeared in an Array

Problem 2: First Missing Positive

Problem 3: Find All Duplicates in an Array

Section 8 — Merge Intervals

What is the Merge Intervals Pattern?

- **The Core Idea:** Process overlapping intervals by sorting them and merging adjacent ranges.
- **Why it works:** Once sorted by start time, overlapping intervals will be adjacent. We can merge them in a single linear pass.
- **Common Triggers:**
  - Finding overlapping ranges, free slots, or busy schedules.
  - Inserting new intervals into already sorted ranges.
  - Finding intersections of multiple lists of intervals.

Merging Logic

Given two sorted intervals A and B, they overlap if:

$$B.\text{start} \leq A.\text{end}$$

If they overlap, we merge them into a single interval C where:

$$C.\text{start} = A.\text{start}$$

$$C.\text{end} = \max(A.\text{end}, B.\text{end})$$

Otherwise, they do not overlap, and we add A to our output list and set B as the new active interval.

Problem 1: Merge Intervals

Problem 2: Insert Interval

Problem 3: Employee Free Time

Section 9 — Modified Binary Search

What is Modified Binary Search?

- **The Core Idea:** An extension of standard binary search that applies to non-standard sorted structures or search spaces.
- **Why it works:** By dividing the search space in half in $O(1)$ time, we achieve a logarithmic $O(\log N)$ runtime.
- **Common Triggers:**
  - Searching in a rotated sorted array.
  - Finding boundaries, peaks, or transition points.
  - **Binary Search on Answers:** When checking if a candidate answer is valid takes $O(N)$ time, and the answer is monotonic, we search the answer range.

Binary Search on Answer Space

Instead of searching indices in an array, we search value ranges (from ‘minPossibleAnswer‘ to ‘maxPossibleAnswer‘):

- Find the midpoint ‘mid‘ of the current range.
- Run a validation function ‘isValid(mid)‘ to check if ‘mid‘ is a possible solution (usually in $O(N)$ time).
- Based on the result:
  - If valid: Try to find a better solution in the lower half (if minimizing).
  - If invalid: Discard the lower half and search the upper half.

Problem 1: Search in Rotated Sorted Array

Problem 2: Find Minimum in Rotated Sorted Array

Problem 3: Capacity To Ship Packages Within D Days

Problem 4: Pow(x, n)

Section 10 — Fast & Slow Pointers

What is the Fast & Slow Pointers Pattern?

- **The Core Idea:** Use two pointers traversing a linear data structure (linked list/array) at different speeds.
- **Why it works:** If there is a cycle, the fast pointer (moving 2 steps at a time) will eventually meet the slow pointer (moving 1 step at a time) in $O(N)$ time and $O(1)$ space.
- **Common Triggers:**
  - Cycle detection in a linked list.
  - Finding the starting node of a cycle.
  - Finding the middle node of a list (when the fast pointer reaches the end, the slow pointer is in the middle).
  - Identifying duplicates in structures where values can be treated as indices.

Cycle Detection Mechanics

- **Initialize:** ‘slow = head‘, ‘fast = head‘.
- **Advance:**
  - ‘slow = slow.next‘
  - ‘fast = fast.next.next‘
- **Check:** If ‘slow == fast‘, a cycle is detected! If ‘fast‘ or ‘fast.next‘ is null, there is no cycle.
- **Cycle Start (Floyd’s algorithm):** Once met, reset ‘slow‘ to ‘head‘. Move both pointers 1 step at a time. The point where they meet again is the start of the cycle.

Problem 1: Linked List Cycle

Problem 2: Middle of the Linked List

Problem 3: Find the Duplicate Number

Section 11 — Tree Depth First Search (DFS)

Tree Depth First Search (DFS)

- **The Core Idea:** Explore as deep as possible along each branch of a tree before backtracking.
- **Implementation:** Typically implemented using recursion (which uses the call stack implicitly) or an explicit Stack.
- **Common Triggers:**
  - Finding paths that sum to a target value.
  - Computing properties of nodes recursively (height, balance).
  - Validating BST properties.
  - Exploring combinations/subtrees.

DFS Traversal Types

Depending on when the current node is processed relative to its children:

- **Pre-order (Root, Left, Right):** Processing parent before children. Useful for copying trees or serialization.
- **In-order (Left, Root, Right):** Processing left child, then parent, then right child. Yields sorted keys in a Binary Search Tree (BST).
- **Post-order (Left, Right, Root):** Processing children before parent. Useful for bottom-up computation (e.g., node height, subproblem accumulation).

Problem 1: Path Sum

Problem 2: Path Sum III

Problem 3: Validate Binary Search Tree

Problem 4: Lowest Common Ancestor of a Binary Tree

Problem 5: Binary Tree Maximum Path Sum

Section 12 — Tree Breadth First Search (BFS)

Tree Breadth First Search (BFS)

- **The Core Idea:** Traverse a tree level-by-level, processing all nodes at depth d before any node at depth $d + 1$.
- **Implementation:** Typically implemented iteratively using a Queue (FIFO).
- **Common Triggers:**
  - Level order traversals or printing.
  - Finding the minimum depth of a tree.
  - Tracking "level" boundaries (e.g. connecting next pointers).
  - Finding the shortest path in unweighted graphs or state transitions.

Level Size Strategy

To process nodes level by level rather than as a continuous queue stream:

- At the start of each level iteration, count the queue size: ‘val levelSize = queue.size‘.
- Run a nested loop exactly ‘levelSize‘ times.
- Inside the loop:
  - Pop a node from the queue.
  - Enqueue its children (left, then right).
- Any logic scoped to a level (e.g., calculating level average or reversing order) can be done cleanly.

Problem 1: Binary Tree Level Order Traversal

Problem 2: Binary Tree Zigzag Level Order Traversal

Problem 3: Populating Next Right Pointers in Each Node

Problem 4: Word Ladder

Section 13 — Matrix Traversal (The "Islands" Pattern)

What is the Matrix Traversal Pattern?

- **The Core Idea:** Treat a 2D grid (matrix) as an implicit graph where each cell (r, c) is a node, and its adjacent neighbors (up, down, left, right) are connected by edges.
- **Traversal:** Use Depth First Search (DFS) or Breadth First Search (BFS) to explore connected regions.
- **Common Triggers:**
  - Counting distinct blocks (e.g. islands, provinces).
  - Simulating cell expansion over time (e.g. rotting oranges, fire spreading).
  - Finding paths or reachable regions from boundaries.

Matrix Traversal Mechanics

- **Boundary Check:** Always ensure row ‘r‘ and column ‘c‘ are valid:
$$0 \leq r < R \quad \text{and} \quad 0 \leq c < C$$
- **Directions Array:** Use a utility array to iterate over neighbors cleanly:
$$\text{dirs} = [(-1, 0), (1, 0), (0, -1), (0, 1)]$$
- **Visited Tracking:** Mark grid cells as visited (e.g., set ‘’1‘‘ to ‘’0‘‘ in-place if allowed, or use a ‘visited‘ boolean matrix) to prevent infinite loops.

Problem 1: Number of Islands

Problem 2: Max Area of Island

Problem 3: Rotting Oranges

Problem 4: Pacific Atlantic Water Flow

Section 14 — Union-Find (Disjoint Set)

What is the Union-Find Pattern?

- **The Core Idea:** A data structure that tracks partition of a set into disjoint subsets.
- **Operations:**
  - **Find:** Identify which subset an element belongs to (returns root parent).
  - **Union:** Join two disjoint subsets into a single subset.
- **Common Triggers:**
  - Dynamic connectivity (are nodes X and Y in the same group?).
  - Detecting cycles in undirected graphs.
  - Merging accounts, groups, or equivalence sets.

Optimizations: Path Compression & Union by Rank

Without optimization, a tree can become a line ($O(N)$ lookup).

- **Path Compression:** During ‘find(x)‘, make every node in the search path point directly to the root root. This flattens the tree structure:

$$\text{parent}[x] = \text{find}(\text{parent}[x])$$

- **Union by Rank/Size:** Always attach the smaller tree under the root of the larger tree.
- **Complexity:** Combined, they achieve nearly constant time operations: $O(\alpha(N))$ where α is the Inverse Ackermann function.

Problem 1: Number of Provinces

Problem 2: Redundant Connection

Problem 3: Accounts Merge

Section 15 — Top ‘K’ Elements (Heap / Priority Queue)

What is the Top 'K' Elements Pattern?

- **The Core Idea:** Find the K largest, smallest, or most frequent elements in a large collection.
- **Why it works:** Instead of sorting the entire array in $O(N \log N)$ time, we can maintain a Heap/Priority Queue of size K.
- **Heaps selection rule:**
  - To find the **K largest** elements: Use a **Min-Heap**. When size exceeds K, pop the smallest element. The heap will retain the K largest values.
  - To find the **K smallest** elements: Use a **Max-Heap**. When size exceeds K, pop the largest.
- **Complexity:** $O(N \log K)$ time and $O(K)$ extra space.

Problem 1: Top K Frequent Elements

Problem 2: K Closest Points to Origin

Problem 3: Task Scheduler

Problem 4: Reorganize String

Section 16 — Dynamic Programming

What is Dynamic Programming (DP)?

- **The Core Idea:** Solve complex problems by breaking them down into simpler, overlapping subproblems and caching their results to avoid redundant calculations.
- **Key Properties:**
  - **Overlapping Subproblems:** Solving the same subproblems repeatedly.
  - **Optimal Substructure:** The optimal solution of the problem can be constructed from optimal solutions of its subproblems.
- **Approaches:**
  - **Top-Down (Memoization):** Recursive, start from target, solve subproblems and save values in a cache.
  - **Bottom-Up (Tabulation):** Iterative, start from base cases, fill a table (array/matrix) up to target.

Dynamic Programming Sub-patterns

Most interview DP problems fall into standard subclasses:

- **Fibonacci Numbers:** ‘dp[i] = dp[i-1] + dp[i-2]‘ (Climbing Stairs).
- **0/1 Knapsack:** Items can be chosen at most once. Decide to include or exclude an item: ‘dp[i][w] = max(exclude, include)‘.
- **Unbounded Knapsack / Coin Change:** Items can be chosen multiple times.
- **Longest Common Subsequence (LCS) / Edit Distance:** String matching.
- **Longest Increasing Subsequence (LIS):** Decisions based on previous elements.

Problem 1: Climbing Stairs

Concept: 0/1 Knapsack

Problem 3: Coin Change

Problem 4: Coin Change 2

Problem 5: Longest Increasing Subsequence

Problem 6: Word Break

Section 17 — Backtracking & DFS

What is the Backtracking Pattern?

- **The Core Idea:** A systematic search algorithm that builds candidate solutions incrementally, and abandons a candidate ("backtracks") as soon as it determines the candidate cannot lead to a valid solution.
- **How it works (The Three Steps):**
  1. **Choose:** Add an element to the current path.
  2. **Explore:** Recursively call the search function with the new state.
  3. **Unchoose (Backtrack):** Remove the element from the current path to return to the previous state.
- **Common Triggers:**
  - Generating all permutations, combinations, or subsets.
  - Solving constraint satisfaction puzzles (e.g. N-Queens, Sudoku).
  - Exploring word combinations on a grid.

Problem 1: Generate Parentheses

Problem 2: Combination Sum

Problem 3: Palindrome Partitioning

Problem 4: Word Search

Problem 5: N-Queens

Section 18 — Greedy Algorithms

What are Greedy Algorithms?

- **The Core Idea:** An algorithmic paradigm that builds up a solution piece by piece, always choosing the next piece that offers the most obvious and immediate local benefit ("greedy" choice).
- **Why it works:** For certain problems, making locally optimal decisions leads to a globally optimal solution.
- **Tradeoffs:**
  - Highly efficient (usually $O(N)$ or $O(N \log N)$).
  - Difficult to mathematically prove correctness.
  - If greedy fails, you must fall back to Dynamic Programming (which checks all combinations).
- **Common Triggers:**
  - Scheduling, interval selection, minimizations.
  - Gas stations, container loading, coin distributions (canonical cases).

Problem 1: Gas Station

Problem 2: Candy

Problem 3: Queue Reconstruction by Height

Section 19 — Data Streams & Online Algorithms

Data Streams & Online Algorithms

- **The Core Idea:** Process data that arrives continuously over time, rather than having the entire dataset available in memory upfront.
- **Constraint:** Algorithms must run in real-time with limited space ($O(1)$ or $O(K)$ where K is the query window size) and time complexity per element.
- **Key Techniques:**
  - **Two Heaps:** Balancing a Max-Heap and Min-Heap to find median in $O(1)$ time.
  - **Reservoir Sampling:** Selecting a random sample from a stream of unknown size.
  - **Double Linked List + HashMap:** Designing fast cache policies (like LRU/LFU cache) with $O(1)$ insertion, lookup, and deletion.

Problem 1: Find Median from Data Stream

Problem 2: Random Pick Index

Problem 3: Moving Average from Data Stream

Problem 4: LRU Cache

Section 20 — Advanced Patterns

Advanced Algorithmic Patterns

For Senior and Staff level roles, interviewers often present advanced structures:

- **Segment Tree:** A binary tree used for storing intervals or segments. It allows querying a range (sum, min, max) and modifying elements in $O(\log N)$ time.
- **Topological Sort:** Ordering vertices in a directed acyclic graph (DAG) such that for every directed edge uv, vertex u comes before v. (Kahn's BFS or DFS).
- **Sweep Line:** A geometry paradigm that solves range/interval problems by sorting boundary events (start/end) and sweeping a imaginary line across them.
- **Trie (Prefix Tree):** A search tree used to store associative keys (typically strings) efficiently. Excellent for autocompletion, spelling checkers, prefix matching.

Segment Tree: Range Sum Query - Mutable

Topological Sort: Course Schedule & Course Schedule II

Sweep Line: The Skyline Problem

Tries: Implement Trie & Replace Words

Section 21 — Miscellaneous Techniques

Miscellaneous Interview Techniques

Some interview questions don't fit into single patterns, but require a combination of general skills:

- **In-place Grid Simulation:** Modifying a matrix without using extra memory by using rows/columns headers as state markers (e.g. Set Matrix Zeroes).
- **Math & Coordinate Transformations:** Mirroring, swapping, or transposing grids (e.g. Rotate Image).
- **Stack Evaluation:** Parsing arithmetic strings or expressions sequentially (e.g. Evaluate Reverse Polish Notation).
- **Bitmasking or Row/Column hashing:** Validating board boundaries quickly (e.g. Valid Sudoku).

Problem 1: Set Matrix Zeroes

Problem 2: Valid Sudoku

Problem 3: Rotate Image

Problem 4: Evaluate Reverse Polish Notation

Section 22 — Summary & The Modern Interview Landscape

Systematic Problem-Solving Template

How to tackle any unknown problem in an interview:

1. **Clarify:** Ask questions about constraints (n, value limits, sorted status, memory limits).
2. **Example:** Dry run a custom, non-trivial test case.
3. **Brute Force:** State the obvious solution ($O(N^2)$ or $O(2^N)$), explaining its bottleneck.
4. **Optimize:** Apply pattern matching (e.g. "Do I need a window?", "Can a map help?").
5. **Code:** Write clean, modular code, stating edge cases.
6. **Dry Run:** Trace the code line-by-line with the example.

The Engineering Manager's Lens

Interviews are a signal of how you will perform as a team member:

- **Tradeoffs:** Be vocal about time vs. space complexity tradeoffs.
- **Production-Ready Code:** Favor clean modular helper methods, input validation, and proper typing over dense one-liners.
- **Collaboration:** Treat the interviewer as a teammate. Don't go silent for minutes; explain your thought flow continuously.

Redefining the SDLC in the AI Era

How AI coding agents (like Gemini, Copilot) change the baseline expectation:

- **Syntax is Cheap:** Being a human compiler or memorizing syntax is no longer the primary value.
- **Logic & System Design is King:** Focus on high-level reasoning, system architecture, identifying constraints, and verifying correctness.
- **Coding speed vs. System security:** Writing code quickly is easy; ensuring it is robust, secure, and fits into the existing framework is the real challenge.

Recap & Roadmap Beyond the Course

- You have studied 20+ crucial algorithmic patterns!
- **Next Steps:**
  - Practice spaced repetition (solve a problem, review it after 3 days, then 7 days).
  - Conduct mock whiteboard interviews with peers.
  - Build intuition over syntax.

Congratulations on completing the course!